PMP Exam Formulas based on *PMBOK® Guide – Sixth Edition*

Keywords	Formulas
Schedule Performance Index (SPI)	A) SPI = EV/PV EV = Earned Value PV = Planned Value [*PMBOK® Guide - Sixth Edition - Page* 267] B) < 1 Project is behind schedule = 1 Project is on schedule > 1 Project is ahead of schedule [*PMBOK® Guide - Sixth Edition - Page* 267]
Cost Performance Index (CPI)	A) CPI = EV/AC EV = Earned Value AC = Actual Cost [*PMBOK® Guide - Sixth Edition - Page* 267] B) < 1 Over budget = 1 On budget > 1 Under budget [*PMBOK® Guide - Sixth Edition - Page*
Schedule Variance (SV)	A) SV = EV - PV EV = Earned Value PV = Planned Value [*PMBOK® Guide - Sixth Edition - Page* 267] B) Negative Behind schedule Zero On schedule Positive Ahead of schedule [*PMBOK® Guide - Sixth Edition - Page* 267]
Cost Variance (CV)	A) CV = EV - AC EV = Earned Value AC = Actual Cost [*PMBOK® Guide - Sixth Edition - Page* 267] B) Negative Over budget Zero On budget Positive Under budget [*PMBOK® Guide - Sixth Edition - Page* 267]
Estimate at Completion (EAC)	EAC = (BAC/CPI) expect future spending to continue at previous rate BAC = Budget at completion CPI = Cost performance index [*PMBOK® Guide - Sixth Edition - Page* 267]
Estimate at Completion (EAC) - If future work will be accomplished at the planned rate/use	EAC = AC + (BAC – EV) AC = Actual cost BAC = Budget at completion EV = Earned value [*PMBOK® Guide - Sixth Edition - Page* 267]
Estimate to Complete (ETC)	ETC = EAC-AC ETC = New Reestimate
Variance at Completion (VAC)	ETC = BAC-AC

Estimate at Completion (EAC) - If initial plan is no longer valid	EAC = AC + Bottom-up ETC [*PMBOK® Guide - Sixth Edition - Page* 267]
Estimate at Completion (EAC) - if both the CPI and SPI influence the remaining work	EAC = AC + (BAC – EV) CPI x SPI CPI = Cost performance index SPI = Schedule performance index [*PMBOK® Guide - Sixth Edition - Page* 267]
Number of Communication Channels	N(N-1)/2 Where N = Number of project team members [*PMBOK® Guide - Sixth Edition - Page* 370]
Three-Point Estimating or PERT Estimation	(O+4M+P)/6 O= Optimistic estimate M= Most Likely estimate P= Pessimistic estimate [*PMBOK® Guide - Sixth Edition - Page* 244]
To-Complete Performance Index (TCPI) based on the BAC	TCPI = (BAC – EV) / (BAC – AC) BAC = Budget at completion AC = Actual cost EV = Earned value [*PMBOK® Guide - Sixth Edition - Page* 266]
To-Complete Performance Index (TCPI) based on the EAC	TCPI = (BAC – EV) / (EAC – AC) BAC = Budget at completion AC = Actual cost EV = Earned value EAC = Estimate at completion [*PMBOK® Guide - Sixth Edition - Page* 266]
Total Float (or) Total Slack	LS-ES (or) LF-EF LS = Late start ES = Early start LF = Late finish EF = Early finish
Standard Deviation of a Task	(P-O)/6 P = Pessimistic estimate O = Optimistic estimate
Present Value (PV)	PV = FV/ (1 + r/100)n N = Number of years r = Discount rate
Net Present Value (NPV)	The higher the better
Internal Rate of Return (IRR)	The higher the better
The Payback Period	The lower the better
The Life Cycle Cost	The lower the better
The Benefit to Cost Ratio (BCR)	The higher the better
Critical Path	Path with longest duration
Rough Order of Magnitude (ROM) Estimate	Estimated value + or - 50% [*PMBOK® Guide - Sixth Edition - Page* 241]
Variance	(Standard Deviation) * (Standard Deviation)

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